

CRANWELL, United Kingdom

WMO: 033790

Lat: 53.0311N

Lon: 0.5036W

Elev: 218

StdP: 14.58

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 25.9	(c) 28.2	(d) 21.2	(e) 16.0	(f) 30.7	(g) 23.5	(h) 17.9	(i) 31.4	(j) 37.7	(k) 48.3	(l) 33.9	(m) 48.7	(n) 7.9	(o) 250	(p) 0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.7	(c) 79.4	(d) 65.0	(e) 75.6	(f) 63.4	(g) 72.5	(h) 61.9	(i) 66.6	(j) 75.4	(k) 64.8	(l) 72.8	(m) 63.3	(n) 70.4	(o) 11.3	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 63.5	(b) 88.3	(c) 70.0	(d) 61.8	(e) 83.2	(f) 67.9	(g) 60.3	(h) 78.8	(i) 66.2	(j) 31.3	(k) 75.4	(l) 29.9	(m) 72.8	(n) 28.8	(o) 70.5	(p) 78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 30.5	(b) 27.1	(c) 24.1	(e) 20.5	(f) 86.5	(g) 3.4	(h) 3.4	(i) 18.1	(j) 88.9	(k) 16.1	(l) 90.8	(m) 14.2	(n) 92.7	(o) 11.8	(p) 95.2		
			DB													
			WB													
				20.5	86.5	3.4	3.4	18.1	88.9	16.1	90.8	14.2	92.7	11.8	95.2	
				19.8	71.7	3.3	3.6	17.4	74.2	15.5	76.3	13.6	78.3	11.2	80.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.0	39.4	40.2	43.3	47.5	52.7	58.0	62.4	62.2	57.5	51.5	44.4	40.2
	DBStd	9.76	5.55	5.81	5.42	5.30	5.20	4.71	4.61	4.37	4.19	5.05	5.39	6.28
	HDD50	1492	328	279	216	108	30	1	0	0	1	42	179	308
	HDD65	5538	793	696	673	524	380	215	108	112	229	421	617	770
	CDD50	1493	1	3	8	34	115	242	386	377	225	87	12	3
	CDD65	64	0	0	0	0	0	7	29	24	3	1	0	0
	CDH74	488	0	0	0	1	7	69	224	160	23	4	0	0
	CDH80	83	0	0	0	0	0	11	37	33	2	0	0	0
Wind	WSAvg	11.8	13.7	13.6	12.8	11.4	11.0	10.2	10.0	10.3	10.9	12.0	12.5	13.3
Precipitation	PrecAvg	23.0	2.0	1.4	1.7	1.9	1.8	2.1	1.8	2.4	2.0	1.9	2.1	2.0
	PrecMax	28.6	4.4	5.2	4.1	4.5	4.7	5.7	5.8	5.3	5.4	4.6	4.2	5.7
	PrecMin	15.7	0.4	0.2	0.2	0.3	0.2	0.1	0.4	0.1	0.3	0.2	0.9	0.5
	PrecStd	3.7	1.0	0.9	0.9	1.0	1.1	1.4	1.2	1.3	1.4	1.2	0.9	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	57.0	61.1	69.8	74.2	81.4	84.4	84.2	77.4	68.6	59.9	56.4
		MCWB	51.9	51.9	52.1	57.1	61.9	66.7	66.7	65.7	63.0	60.9	56.6	53.0
	2%	DB	52.3	53.6	57.6	63.9	70.0	75.0	79.6	78.5	72.2	64.0	57.1	54.3
		MCWB	49.2	49.6	50.3	54.0	59.9	62.6	65.2	64.9	62.2	58.2	54.2	51.7
	5%	DB	50.2	51.2	54.8	60.2	66.3	70.8	76.1	74.1	68.3	61.4	54.8	52.0
		MCWB	47.5	47.6	49.2	51.9	57.3	60.7	63.7	63.0	59.9	57.0	52.0	49.6
	10%	DB	48.0	49.1	52.3	57.1	62.8	67.7	72.5	71.0	65.5	59.3	52.6	49.6
		MCWB	45.6	45.9	47.4	50.2	55.0	59.0	61.9	61.4	58.2	55.3	50.2	47.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.3	52.7	54.1	58.5	63.8	68.1	69.9	68.7	66.0	62.0	57.3	53.7
		MCDB	54.7	55.9	58.6	67.2	71.7	78.6	81.6	77.8	73.0	66.9	59.3	55.4
	2%	WB	49.8	50.1	51.9	55.6	60.5	64.3	66.6	66.2	63.2	59.8	54.7	52.0
		MCDB	51.9	53.1	56.0	62.0	68.3	71.8	75.8	74.6	69.7	63.0	56.6	54.1
	5%	WB	47.9	48.2	49.9	53.1	58.3	62.0	64.8	64.5	61.3	57.8	52.5	49.9
		MCDB	49.8	50.8	54.0	58.8	64.7	68.8	73.2	72.1	66.4	60.6	54.3	51.7
	10%	WB	45.8	46.3	48.0	50.9	56.2	60.2	63.1	62.9	59.6	56.0	50.5	47.6
		MCDB	47.7	48.7	51.7	56.2	61.5	66.3	70.6	69.3	63.9	58.6	52.2	49.5
Mean Daily Temperature Range	5% DB	MDBR	9.1	10.7	13.0	15.1	15.6	15.6	16.7	16.0	14.6	12.2	10.0	8.8
		MCDBR	11.6	12.4	16.5	19.7	21.2	21.5	22.8	20.8	18.7	13.9	11.2	10.7
		MCWBR	10.4	9.8	10.7	11.5	12.4	11.3	10.5	9.4	10.4	9.5	9.5	9.9
	5% WB	MCDBR	11.5	12.0	15.0	18.3	19.4	19.5	20.4	18.6	17.0	13.5	10.9	10.6
		MCWBR	10.6	10.1	10.7	11.6	12.3	11.5	10.5	9.3	10.7	10.2	9.8	10.1
Clear-Sky Solar Irradiance	taub		0.323	0.339	0.371	0.397	0.399	0.403	0.404	0.392	0.370	0.351	0.325	0.311
	taud		2.359	2.388	2.294	2.231	2.267	2.277	2.284	2.344	2.406	2.442	2.409	2.335
	Ebn at Noon		207	243	256	264	270	269	266	263	255	235	206	187
	Edh at Noon		19	24	33	39	40	40	39	35	29	23	18	16
All-Sky Solar Radiation	RadAvg		225	442	782	1204	1501	1577	1527	1258	935	539	277	181
	RadStd		21	52	94	139	98	152	163	85	85	46	31	19

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+76	N/A
Regional (82 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page